

# Literature Status for the Main Conjecture of arXiv:1606.06913

Automated mathematics run `fa_banach_001`

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## Status

**Result type:** literature already answered.

**Source:** Daniel Dadush and Oded Regev, *Towards Strong Reverse Minkowski-type Inequalities for Lattices*, arXiv:1606.06913.

**Answer:** Oded Regev and Noah Stephens-Davidowitz, *A Reverse Minkowski Theorem*, arXiv:1611.05979.

## Source conjecture

Conjecture 3.1 (“Main conjecture”) of the source, on PDF page 14, defines  $C_\eta(n)$  to be the least constant for which every full-rank lattice  $\mathcal{L} \subset \mathbb{R}^n$  satisfies

$$\eta(\mathcal{L}) \leq C_\eta(n) \max_{\substack{W \neq \{0\} \\ W \text{ a lattice subspace of } \mathcal{L}^*}} \det(\mathcal{L}^* \cap W)^{-1/\dim W}, \quad (1)$$

and conjectures

$$C_\eta(n) \leq \text{polylog } n.$$

The source immediately notes the homogeneous normalization: it suffices to prove  $\eta(\mathcal{L}) \leq \text{polylog } n$  whenever every sublattice of  $\mathcal{L}^*$  has determinant at least one.

Here  $\eta(\mathcal{L})$  is the  $1/2$ -smoothing parameter, characterized by

$$\sum_{y \in \mathcal{L}^*} e^{-\pi s^2 \|y\|^2} = \frac{3}{2} \quad \text{at } s = \eta(\mathcal{L}).$$

## Later theorem

The abstract and introduction of Regev–Stephens-Davidowitz explicitly say that their main result proves Dadush’s conjecture and cite the source paper. Their Theorem 1.2 (“Reverse Minkowski Theorem”), on PDF page 3, states that if  $\Gamma \subset \mathbb{R}^n$  is a lattice satisfying

$$\det(\Gamma') \geq 1 \quad \text{for every sublattice } \Gamma' \subseteq \Gamma,$$

then

$$\sum_{y \in \Gamma} e^{-\pi t^2 \|y\|^2} \leq \frac{3}{2}, \quad t = 10(\log n + 2). \quad (2)$$

## Exact implication

Set  $\Gamma = \mathcal{L}^*$ . Under the source’s normalized hypothesis, the later theorem is exactly the smoothing inequality that yields

$$\eta(\mathcal{L}) \leq 10(\log n + 2).$$

The later paper also records the homogeneous form explicitly. If

$$\eta_{\det}(\Gamma) = \max_{\Gamma' \subseteq \Gamma} \det(\Gamma')^{-1/\text{rank}(\Gamma')},$$

then its equation (1.3) gives

$$\eta^*(\Gamma) \leq 10(\log n + 2)\eta_{\det}(\Gamma),$$

where  $\eta^*(\mathcal{L}^*) = \eta(\mathcal{L})$  in the source convention. Comparing this with the displayed source inequality proves the following.

**Theorem.** *The Main Conjecture of Dadush–Regev is answered affirmatively in the later literature, with the explicit estimate*

$$C_\eta(n) \leq 10(\log n + 2).$$

This is stronger than the conjectured unspecified polylogarithmic bound.

## Scope and verification

This packet records a known answer rather than a new mathematical result. It resolves the source’s central Main Conjecture; it does not assert that every stronger variant or related question discussed in the source is settled.

The stored source PDF has 65 pages and the supporting PDF has 41 pages. The source statement was checked in Conjecture 3.1, PDF page 14. The answer was checked in Theorem 1.2, PDF page 3, and in the later paper’s equation (1.3), which supplies the scaled formulation used above. Cheap run-index search found no prior packet for this exact identification.

## References

- [1] D. Dadush and O. Regev, *Towards Strong Reverse Minkowski-type Inequalities for Lattices*, arXiv:1606.06913 (2016).
- [2] O. Regev and N. Stephens-Davidowitz, *A Reverse Minkowski Theorem*, arXiv:1611.05979; Ann. of Math. 185 (2017), 233–273.